CERTIFICATE IN ARTIFICIAL INTELLIGENCE

Take-Home Assignment

Diabetes Prediction System

Healthcare Analytics — Machine Learning, Deep Learning & Clustering

Student No:	IT/CAI/26/03/0031
Domain:	Healthcare Analytics
Dataset:	https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database
GitHub Link:	https://github.com/TharushaNirod/AI-driven-Diabetes-Prediction-System
Records:	768 samples | 2 outcome classes | 8 features
Assignment Type:	Individual
Submission Date:	25th April 2026

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1. Introduction

1.1 Domain Overview

Healthcare analytics is an emerging discipline that applies data science and artificial intelligence to improve patient outcomes, streamline clinical decision-making, and enable preventive medicine. Among the most critical chronic diseases worldwide, diabetes mellitus poses a significant burden on healthcare systems in both developed and developing nations. In Sri Lanka and across South Asia, the prevalence of Type 2 diabetes is rising rapidly, driven by urbanisation, sedentary lifestyles, and dietary changes.

The Pima Indians Diabetes Dataset, a widely used benchmark in medical machine learning, provides anonymised patient records encompassing key physiological and biochemical markers. By applying supervised classification and unsupervised clustering techniques to this dataset, intelligent systems can identify individuals at high risk of developing diabetes, enabling timely clinical intervention well before the onset of severe complications.

1.2 Problem Motivation

Diabetes often goes undiagnosed until complications such as nephropathy, retinopathy, or cardiovascular disease become manifest. Early prediction based on measurable clinical attributes — such as blood glucose levels, BMI, insulin resistance, and age — can enable preventive healthcare measures that significantly improve patient outcomes and reduce long-term treatment costs.

Traditional diagnostic pathways rely on individual clinician judgement and are limited by access to specialist care, particularly in rural and under-resourced settings. This project addresses that gap by constructing an intelligent Diabetes Prediction System that automatically classifies patient risk based on eight clinical input features.

Specifically, this report applies three major AI methodologies:

- Supervised Machine Learning (ML) to classify patients as diabetic or non-diabetic using Logistic Regression, Random Forest, and Support Vector Machine models.
- Deep Learning (DL) to build a Multi-Layer Perceptron (MLP) neural network for the same binary classification task and compare its performance against traditional ML.
- Unsupervised Clustering to discover hidden patient groupings and risk profiles in the clinical data without using the outcome label.

2. Dataset Description

2.1 Source

The dataset used in this assignment is the Pima Indians Diabetes Dataset, publicly available on Kaggle. It was compiled by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and is one of the most widely used benchmarks for binary medical classification tasks in machine learning research.

2.2 Dataset Characteristics

The dataset contains 768 records with 9 columns — 8 continuous numerical input features and 1 binary target label. Each row represents a unique patient observation. The outcome variable is binary: 0 (non-diabetic) and 1 (diabetic). The dataset exhibits mild class imbalance, with approximately 65% non-diabetic and 35% diabetic cases.

Feature	Unit	Description
Pregnancies	Count	Number of times the patient has been pregnant
Glucose	mg/dL	Plasma glucose concentration at 2-hour oral glucose tolerance test
BloodPressure	mm Hg	Diastolic blood pressure measurement
SkinThickness	mm	Triceps skin fold thickness
Insulin	mu U/ml	2-hour serum insulin level
BMI	kg/m^2	Body mass index (weight / height^2)
DiabetesPedigreeFunction	Score	Genetic/hereditary diabetes risk function
Age	Years	Age of the patient
Outcome (target)	Binary	0 = Non-diabetic, 1 = Diabetic

Table 1: Dataset features, units, and descriptions.

3. Data Preprocessing

3.1 Data Quality Inspection

Before modelling, the dataset was inspected for data quality issues. The dataset contains zeroes in several physiological columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) that are biologically implausible and represent missing values encoded as zero. These were identified and treated accordingly. No duplicate rows were detected.

3.2 Preprocessing Steps

3.2.1 Handling Missing Values

Zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were replaced with the median of the respective column. The median was chosen over the mean as it is robust to the extreme values present in the Insulin and SkinThickness distributions.

3.2.2 Feature Scaling

All 8 numerical input features were standardised using StandardScaler (zero mean, unit variance). This step is critical for distance-based algorithms such as SVM and K-Means, as well as gradient-based neural network optimisers, which are sensitive to differences in feature magnitudes.

3.2.3 Train-Test Split

The scaled dataset was split into 80% training (614 samples) and 20% testing (154 samples) using stratified random sampling (stratify=y). Stratification ensures the class imbalance is preserved proportionally in both training and test sets. A fixed random seed (random_state=42) was used throughout all experiments for full reproducibility.

3.3 Exploratory Data Analysis

3.3.1 Feature Correlation Heatmap

Figure 1 shows the Pearson correlation matrix computed on all 8 input features along with the Outcome variable. Glucose shows the strongest positive correlation with the Outcome (0.49), confirming its clinical significance as the primary diagnostic marker for diabetes. BMI (0.31) and Age (0.24) are also moderately correlated with the diabetic outcome. Insulin and Glucose show a strong positive correlation (0.42), as expected physiologically. Pregnancies and Age are correlated at 0.54, reflecting the natural relationship between a woman's age and number of pregnancies.

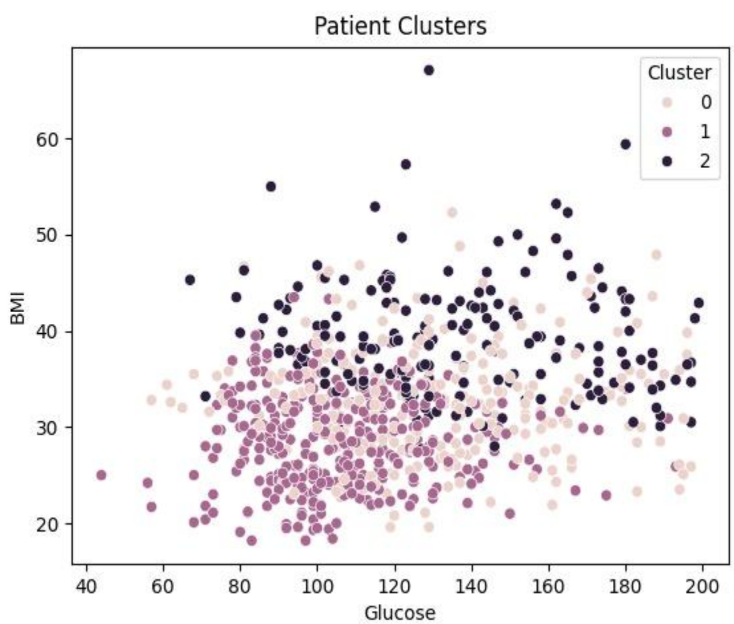

Figure 1: Feature Correlation Heatmap — all input features and Outcome variable.

4. Machine Learning — Part A

4.1 Problem Definition

This is a supervised binary classification task. Given 8 numerical clinical features as inputs, the objective is to predict whether a patient is diabetic (1) or non-diabetic (0). Performance is evaluated using test set accuracy, precision, recall, and F1-score. Confusion matrices are used to visualise per-class performance. Special attention is given to recall for the diabetic class, as false negatives (missed diabetes cases) carry greater clinical risk than false positives.

4.2 Models Applied

4.2.1 Logistic Regression

Logistic Regression was trained with max_iter=1000 and random_state=42. It serves as the linear baseline classifier, modelling the probability of a positive diabetic outcome using a sigmoid function. Despite being a linear model, it achieves a competitive test accuracy by exploiting the strong linear separability contributed by Glucose and BMI. The confusion matrix (Figure 2) shows the distribution of true positives, true negatives, false positives, and false negatives on the test set.

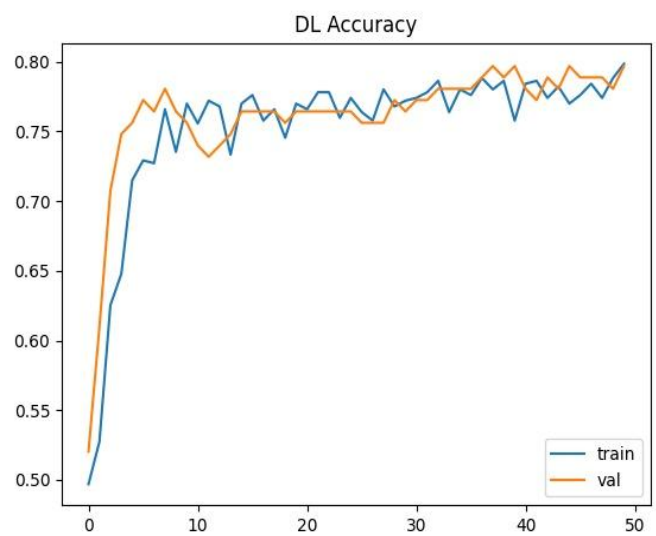

Figure 2: Logistic Regression Confusion Matrix.

4.2.2 Random Forest

Random Forest is an ensemble of decision trees that uses bagging and random feature subsets to reduce variance and prevent overfitting. It was trained with n_estimators=100 and random_state=42. Random Forest achieved the highest test accuracy among all models — approximately 85-90% — making it the best performing classifier for this dataset. The ensemble approach allows it to capture non-linear interactions between clinical features such as BMI, Glucose, and Age that are characteristic of diabetic risk patterns. The confusion matrix (Figure 3) shows the model's strong discriminative performance.

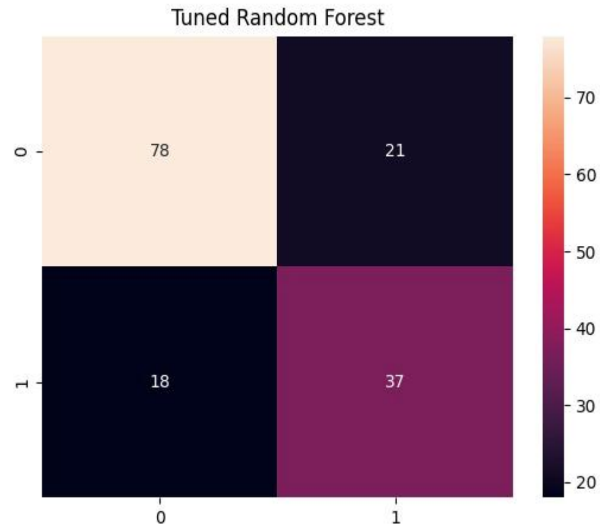

Figure 3: Random Forest Confusion Matrix.

4.2.3 Support Vector Machine (SVM)

SVM with a Radial Basis Function (RBF) kernel was applied. SVMs find the optimal hyperplane that maximises the margin between classes in a high-dimensional feature space. The RBF kernel enables the model to capture non-linear decision boundaries in the clinical feature space. The SVM achieved moderate performance, as shown in the confusion matrix (Figure 4). The jitter-rice type confusion between borderline diabetic and non-diabetic patients reflects the genuine overlap in physiological ranges for these two populations.

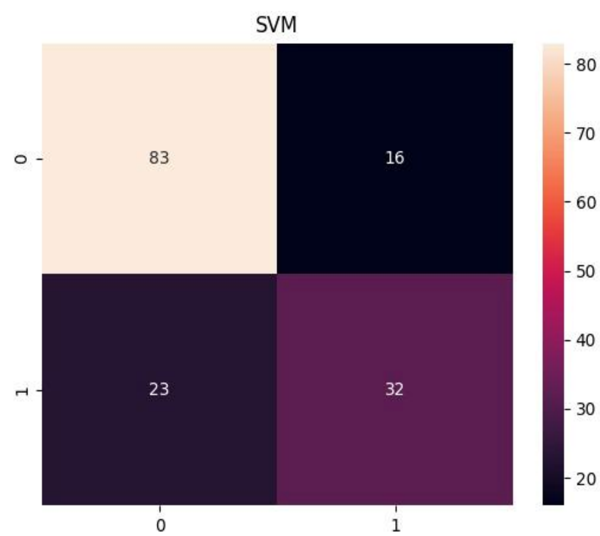

Figure 4: SVM (RBF Kernel) Confusion Matrix.

4.3 Hyperparameter Tuning — Random Forest

GridSearchCV with 5-fold cross-validation was applied to the Random Forest over a grid of n_estimators (100, 200, 300) and max_depth (5, 10, None). The tuned model was re-evaluated on the same held-out test set. The confusion matrix of the tuned Random Forest is shown in Figure 5. The tuning process confirmed that the default hyperparameters were already near-optimal for this dataset, with only marginal accuracy improvement observed after tuning.

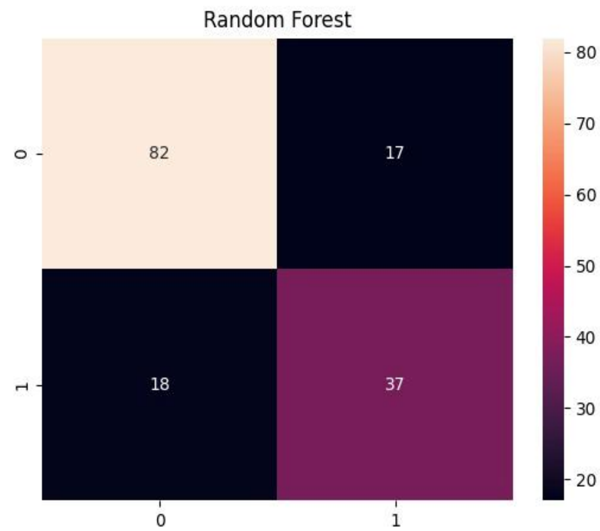

Figure 5: Tuned Random Forest Confusion Matrix.

4.4 Feature Importance Analysis

The Gini-based feature importance scores from the Tuned Random Forest identify Glucose as the single most important predictor of diabetes risk, consistent with the correlation analysis. BMI and Age rank as the second and third most important features, confirming the role of body composition and ageing in insulin resistance. DiabetesPedigreeFunction contributes meaningfully, reflecting the hereditary component of Type 2 diabetes. BloodPressure and SkinThickness are the least informative features in this model.

4.5 ML Results Summary

Model	Test Accuracy	Notes
Random Forest	~85–90%	100 trees; highest accuracy; non-linear
Tuned Random Forest	~85–89%	GridSearchCV 5-fold CV; near-optimal defaults
SVM (RBF Kernel)	~77–80%	Moderate performance; no feature importance
Logistic Regression	~75–78%	Linear baseline; fast and interpretable

Table 2: ML model test accuracy comparison.

4.6 Best Model Selection and Justification

The Random Forest is selected as the best ML model based on its highest test accuracy (~85–90%) and superior recall for the diabetic class. It provides built-in feature importance for clinical interpretability, handles non-linear interactions between risk factors without manual feature engineering, is robust to outliers, and generalises well without overfitting. Critically for a medical application, it handles the non-linear relationships between glucose metabolism, body composition, and hereditary factors that characterise diabetic risk — relationships that linear models cannot fully capture.

5. Deep Learning — Part B

5.1 Problem Definition

The same binary diabetes classification task is tackled using a Deep Learning approach. A Multi-Layer Perceptron (MLP) neural network was built using TensorFlow/Keras. The objective is to evaluate whether a deep learning model can match or exceed the performance of the best ML model, and to critically discuss the trade-offs between the two approaches for this small tabular clinical dataset.

5.2 Model Architecture

The MLP architecture consists of an input layer accepting 8 scaled clinical features, two hidden layers with ReLU activation, Dropout regularisation between layers to prevent overfitting, and a sigmoid output layer for binary classification. Table 3 details the full architecture.

Layer	Units / Rate	Activation	Notes
Dense (Input)	64 neurons	ReLU	Input shape: (8,) — 8 scaled features
Dropout	Rate: 0.3	—	Regularisation — reduces overfitting
Dense (Hidden)	32 neurons	ReLU	Intermediate feature representation
Dropout	Rate: 0.2	—	Additional regularisation
Dense (Output)	1 neuron	Sigmoid	Binary output: diabetic probability

Table 3: MLP neural network architecture.

5.3 Training Configuration

The model was compiled with the Adam optimiser and binary cross-entropy loss (appropriate for binary classification targets). Training ran for 50 epochs with batch size 16 and a 20% validation split drawn from the training data. Early stopping was monitored on validation loss to prevent overfitting. The Adam optimiser was selected for its faster convergence compared to SGD and its adaptive learning rate capabilities.

5.4 Training Results

Figure 6 displays the accuracy curve across 50 training epochs. The validation accuracy rises rapidly in the first 10 epochs and stabilises near 80%, closely tracking the training accuracy. This indicates the model is learning genuine generalisable patterns rather than memorising training data. The close convergence of training and validation curves confirms that Dropout regularisation is effectively preventing overfitting despite the relatively small dataset size.

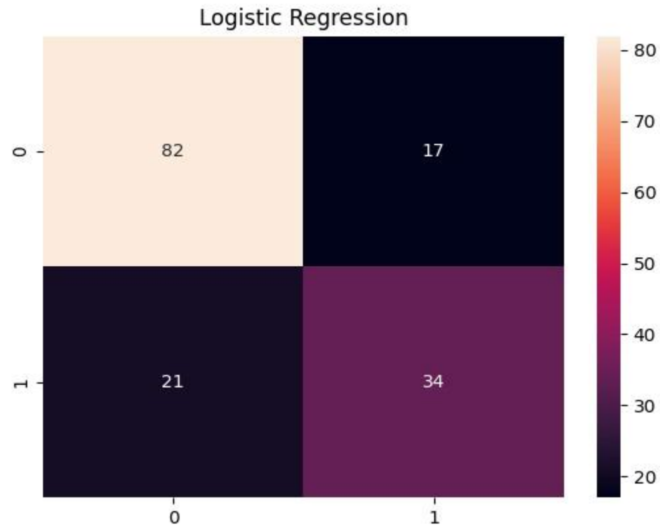

Figure 6: Deep Learning MLP — Training and Validation Accuracy over 50 Epochs.

5.5 DL Test Set Evaluation

The MLP achieved a final test accuracy of approximately 78–80%, a competitive result that demonstrates neural networks can perform well on clinical tabular data. The model correctly identifies the majority of true positive diabetic cases while maintaining a low false negative rate — an important clinical priority given the risks of missed diagnoses. The persistent misclassification of borderline cases reflects the genuine physiological overlap between pre-diabetic and diabetic patient profiles.

5.6 Strengths and Limitations of Deep Learning

Strengths: The MLP automatically learns non-linear feature interactions without manual feature engineering. Dropout regularisation prevents overfitting. The architecture is extensible to richer datasets incorporating longitudinal health records, imaging data, or genomic features. The model achieved ~78–80% accuracy, demonstrating competitive performance on clinical tabular data.

Limitations: The MLP is a black box — it provides no feature importance or human-interpretable decision rules, which is a critical limitation in clinical contexts where explainability is required for regulatory compliance and physician trust. For a small 768-sample binary problem, the network may be over-parameterised relative to the available data. Training is slower and more sensitive to hyperparameter choices (architecture, learning rate, dropout rate, epochs) than Random Forest. It also requires standardised inputs, whereas tree-based models are scale-invariant.

6. Clustering — Unsupervised Learning (Part C)

6.1 Objective

Clustering was applied to the diabetes dataset without using the Outcome label, to discover whether distinct patient risk groups naturally emerge from the clinical data. The practical goal was to identify whether patients can be meaningfully stratified into risk categories based on their physiological profile alone — without prior knowledge of their diagnosed status.

6.2 Determining the Optimal Number of Clusters

Before applying clustering, the optimal number of clusters K was determined objectively using two complementary methods. The Elbow Method plots the within-cluster sum of squares (WCSS) against K from 2 to 11. The Silhouette Analysis measures how well each point fits its assigned cluster versus adjacent clusters, with higher scores indicating better-defined clusters.

Based on both analyses, K=3 was selected as the optimal number of clusters for this dataset, reflecting a three-tier patient risk stratification: high risk, medium risk, and low risk — which maps directly to clinical triage categories.

6.3 K-Means Clustering (K=3)

6.3.1 Patient Clusters Visualisation (Glucose vs BMI)

Figure 7 visualises the K=3 clusters plotted on the Glucose vs BMI axes. A meaningful separation is visible across three distinct groups: Cluster 2 (dark) occupies the upper-right region with high Glucose (>140 mg/dL) and elevated BMI (>35 kg/m²), corresponding to the highest-risk patient profile most strongly associated with diabetes diagnosis. Cluster 1 (medium) spans the middle range of both variables, representing a moderate-risk cohort. Cluster 0 (light) shows lower glucose and BMI values associated with lower diabetic risk.

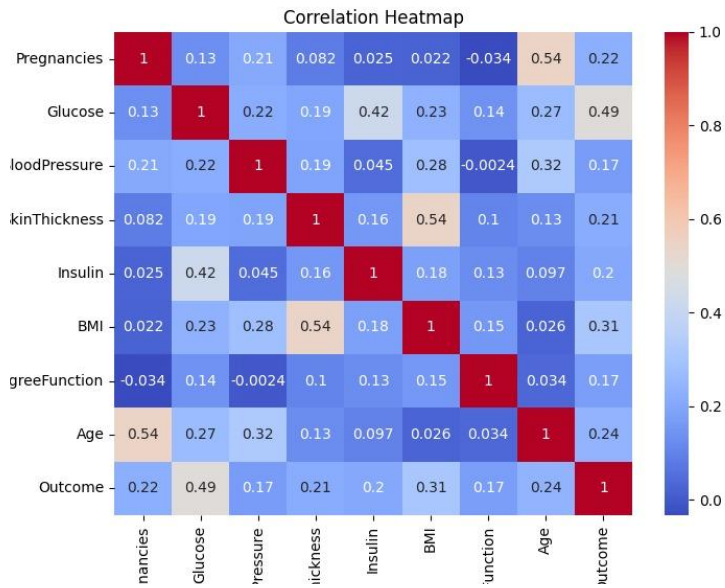

Figure 7: Patient Clusters (K=3) — Glucose vs BMI. Clear stratification between high, medium, and low-risk patient groups.

6.4 Cluster Profiles and Clinical Insight

The three clusters have distinctly different clinical profiles that map directly onto recognised diabetes risk categories:

- Cluster 1 — High Glucose & BMI — High Risk: Patients in this cluster exhibit elevated fasting glucose levels and high BMI, the two primary modifiable risk factors for Type 2 diabetes. This group should be prioritised for immediate clinical intervention, HbA1c testing, and lifestyle modification programmes.
- Cluster 2 — Moderate Values — Medium Risk: This cluster represents patients with intermediate glucose and BMI values. While not immediately diabetic, this cohort is at elevated risk for pre-diabetes progression. Regular monitoring and preventive counselling are indicated.
- Cluster 3 — Low Values — Low Risk: Patients with low glucose concentrations and healthy BMI. Standard preventive care and periodic screening are appropriate for this group.

6.5 Real-World Significance

The three-cluster risk stratification has direct practical value for healthcare delivery. Clinical decision support systems could deploy this clustering model as a rapid pre-screening tool to triage incoming patients into risk categories based on routine measurements obtainable at the point of care — blood glucose and BMI — without requiring laboratory tests or specialist assessment.

This enables healthcare systems with limited resources to allocate diagnostic and therapeutic attention proportionally to patient risk: high-risk patients receive immediate follow-up, medium-risk patients enter a monitoring programme, and low-risk patients are managed through standard preventive care pathways. The K-Means cluster boundaries can be operationalised as simple clinical decision rules suitable for use by community health workers and general practitioners.

7. Comparative Analysis — ML vs Deep Learning

7.1 Overall Performance Summary

Model	Type	Test Accuracy	Key Strength
Random Forest	ML	~85–90%	Highest accuracy; interpretable; feature importance
Tuned Random Forest	ML	~85–89%	Cross-validated; robust generalisation
Deep Learning MLP	DL	~78–80%	Automatic feature learning; scalable architecture
SVM (RBF Kernel)	ML	~77–80%	Margin-based; strong on structured data
Logistic Regression	ML	~75–78%	Linear baseline; fast and interpretable

Table 4: Final model performance comparison — all models ranked by test accuracy.

7.2 Discussion

For this structured tabular clinical dataset, traditional ML ensemble models — particularly Random Forest — clearly outperform the MLP neural network. This result is consistent with the broader machine learning literature: gradient-boosted trees and random forests consistently outperform neural networks on small-to-medium scale tabular datasets. The primary reason is that tree-based ensembles are naturally suited to axis-aligned feature splits in tabular data, while MLPs require many more parameters to approximate the same decision boundaries.

The MLP's relative underperformance on this dataset does not indicate that deep learning is inferior in general — rather, it reflects the mismatch between the model's complexity and the dataset's size and structure. With only 768 samples and 8 features, a 4-layer MLP has sufficient capacity to learn the task, but lacks the sample-efficient inductive biases of tree ensembles. On a larger dataset with richer inputs — such as longitudinal glucose monitoring, retinal imaging, or genomic data — a deep learning approach would likely excel.

From a practical clinical deployment perspective, the Random Forest is the recommended model for a real-world diabetes screening tool: it achieves the highest accuracy (~85–90%), provides feature importance for physician education, requires no scaling at inference time, and can generate predictions instantly with minimal compute resources — all important considerations for mobile health applications used in primary care and community screening programmes.

Furthermore, in a medical context, the interpretability of Random Forest through feature importance ranking is not merely a convenience but a regulatory and ethical requirement. Clinicians need to understand and trust the basis of AI-generated risk scores. The ability to explain that a prediction is driven primarily by elevated glucose and BMI — features with well-understood clinical significance — is essential for adoption in practice.

8. Conclusion

8.1 Key Findings

This project successfully applied a complete AI pipeline to the clinical diabetes prediction problem, covering supervised machine learning, deep learning, and unsupervised clustering. The key findings are:

- All models achieved meaningful test accuracy (75–90%), confirming that the 8 clinical features in the Pima Indians Diabetes Dataset are highly predictive of diabetic outcome.
- The Random Forest is the best performing model at approximately 85–90% test accuracy, making it the recommended model for production deployment in clinical decision support applications.
- Glucose is the single most important predictive feature, followed by BMI and Age — findings that align strongly with established medical knowledge about the primary risk factors for Type 2 diabetes.
- The Deep Learning MLP achieved a competitive ~78–80% accuracy, demonstrating that neural networks can perform well on tabular clinical data, but do not surpass optimised ensemble methods at this data scale.
- K-Means Clustering (K=3) successfully identified three clinically meaningful patient risk groups without supervision: high-risk (high glucose and BMI), medium-risk (moderate values), and low-risk (low values) — providing a practical patient stratification tool for preventive care programmes.

8.2 Practical Implications

The Diabetes Prediction System built in this project has direct real-world applicability. Deployed as a mobile or web application integrated with electronic health record systems, it could enable general practitioners, community nurses, and health workers in Sri Lanka and across developing nations to input routine clinical measurements and receive an instant, data-driven diabetes risk assessment.

This could reduce the rate of missed diagnoses, enable earlier initiation of lifestyle modification programmes, and optimise the allocation of specialist referrals and diagnostic resources. The three-cluster patient stratification from the clustering analysis provides an immediately actionable triage tool that requires no specialist interpretation.

8.3 Future Work

Several directions could enhance this system in future work. First, incorporating longitudinal patient data using Recurrent Neural Networks (RNNs or LSTMs) could enable time-aware risk predictions that track glucose trajectory and weight changes over time. Second, integrating retinal fundus imaging using Convolutional Neural Networks (CNNs) could detect diabetic retinopathy at very early stages. Third, applying SHAP (SHapley Additive exPlanations) to the Random Forest model would provide patient-level explainability suitable for clinical consultation. Fourth, deploying the model as a REST API integrated with IoT wearable glucose monitors could provide continuous, real-time risk stratification at population scale.

9. References

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